Project: Study Coach AI

Week 3 Check-in: Users and Data

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1. Fictional User Personas

Persona 1: Mariam Hassan — Below Typical Technical Level

Background: Mariam is a fictional 19-year-old university student from rural Egypt. She primarily uses her smartphone for studying and has limited experience with AI-powered educational tools.

Technical level: Below typical.

AI experience: Novice. She has tried an AI chatbot once or twice but does not know how to write effective prompts or evaluate AI-generated answers.

Goals: Understand difficult course concepts, prepare for examinations, and practice solving questions independently.

Challenges and pain points:

- Academic terminology and long explanations can be difficult to understand.
- She may not know how to formulate a clear question.
- She may assume that an AI-generated answer is always correct.

Accessibility and interaction needs: Simple language, readable text, short explanations, and clear instructions.

Expected Study Coach AI support: The system should allow Mariam to request hints, simple explanations, and step-by-step guidance. It should identify relevant course sources whenever possible.

Persona 2: Omar Khalid — Typical Technical Level

Background: Omar is a fictional 21-year-old university student from Jordan. He regularly uses learning management systems, online educational resources, and productivity applications.

Technical level: Typical.

AI experience: Novice. Although he is comfortable with digital tools, he has limited experience evaluating the reliability of AI-generated educational content.

Goals: Review course material efficiently, identify mistakes, and prepare for examinations.

Challenges and pain points:

- AI answers may contain information that is not included in the course material.
- He may struggle to distinguish verified information from unsupported explanations.
- Generic answers may not address the exact question he asked.

Accessibility and interaction needs: Clear source references, consistent navigation, and understandable explanations.

Expected Study Coach AI support: The system should connect answers to the approved knowledge source, explain important concepts, and allow Omar to report incorrect or unhelpful answers.

Persona 3: Amina Ahmed— Advanced Technical Level

Background: Amina is a fictional 22-year-old university student from Nigeria. She is experienced with digital learning platforms and AI-assisted study tools.

Technical level: Advanced.

AI experience: Experienced. She understands how to formulate detailed prompts and compare alternative explanations.

Accessibility need: Amina has low vision and benefits from adjustable text size, strong visual contrast, and compatibility with screen readers.

Goals: Explore complex topics, compare explanations, and control the level of detail in study materials.

Challenges and pain points:

- Basic explanations may not provide sufficient depth.
- Small text and poorly structured interfaces can make information difficult to access.
- She may have difficulty verifying an answer when the source is not clearly identified.

Expected Study Coach AI support: The system should provide adjustable explanation levels, structured answers, identifiable sources, and an accessible interface that supports her preferred assistive technology.

Persona Design Rationale

These personas represent different technical skill levels, educational needs, AI experiences, and cultural backgrounds. Mariam and Omar are novice AI users, while Amina represents an advanced user. Amina's low-vision accessibility need highlights the importance of inclusive interaction design.

All three personas are fictional and are not based on identifiable real students.

2. Bias-Check Notes

Purpose

The team reviewed the initial persona descriptions and proposed interaction design for Study Coach AI to identify possible stereotypes, accessibility barriers, and exclusions. The review focuses on language, cultural representation, technical experience, accessibility, and the reliability of educational answers.

Bias Review Table

Area reviewed	Potential bias or exclusion	Proposed change
Names and cultural backgrounds	Assuming that students from a particular culture have the same abilities or learning preferences.	Use culturally varied fictional personas without assigning technical ability based on nationality or culture.
Technical proficiency	Assuming that all university students are equally comfortable with AI tools.	Provide clear instructions, simple prompts, and optional advanced controls.
Language and explanations	Assuming that all students understand complex academic language.	Offer plain-language explanations, definitions, and step-by-step guidance.
Accessibility	Assuming that all users can read small text or interact with a mouse.	Design for adjustable text size, sufficient contrast, keyboard navigation, and screen-reader compatibility.
Educational examples	Using examples that reflect only one cultural or social context.	Review examples for cultural relevance and use varied, inclusive contexts.
AI answer reliability	Presenting generated answers as correct without verification.	Ground answers in the approved knowledge source and provide source references where possible.
User control	Forcing every student to use the same explanation style.	Allow users to request hints, step-by-step explanations, or detailed answers.

Review Findings

The initial design review identified several potential risks. A single explanation style may not serve both novice and advanced users. An interface designed only for sighted users may

exclude students with low vision. Cultural assumptions in examples may also make some students feel that the system was not designed for them.

The team therefore proposes flexible explanation levels, more inclusive educational examples, source-aware answers, and accessibility support.

Planned Validation

Before claiming that these changes are effective, the team will review representative prompts and generated responses, inspect the interface for accessibility barriers, and test whether the explanation controls behave as intended.

The review will use fictional scenarios and synthetic educational examples. No real student records or personally identifiable information will be submitted to AI tools.

Status

This is an initial bias review. Proposed changes will be marked as implemented only after the team has built and tested them.

3. Data Source Description

Proposed Data Source: Synthetic Educational Question-and-Answer Dataset

Study Coach AI is designed to help students understand course materials, practice questions, and prepare for examinations. For the initial prototype, the project uses an AI-assisted synthetic educational question-and-answer dataset containing fictional questions, draft reference answers, short explanations, and topic identifiers.

Dataset Structure

The current dataset contains 30 synthetic question-and-answer records. Each record includes the following fields:

- `question`: A fictional educational question related to Human-AI Interaction (HAI).
- `reference_answer`: A draft answer intended to represent the expected response.
- `explanation`: A short explanation to support student understanding.
- `topic`: The relevant HAI concept.
- `source_reference`: A note indicating the intended source or verification requirement.
- `source_status`: The current verification status of the record.

Dataset Size and Status

The dataset currently contains 30 draft records prepared for initial prototyping. However, these records have not yet been fully verified against the approved HAI course materials. Therefore, the dataset is considered an initial synthetic draft, and its answers must be reviewed and corrected where necessary before it is used as a trusted source of course-specific knowledge.

License and Permission

The dataset consists of AI-assisted synthetic examples prepared for this project and does not intentionally include real students' personal information. No external dataset license is being claimed for these synthetic records. The team will document any external sources used to verify or revise the answers and will follow the relevant access, attribution, and usage requirements. The dataset's final usage terms will be clarified before the project is distributed or published.

Why This Dataset Fits the Project

The dataset is suitable for the initial educational-assistance prototype because it provides a small, controlled collection of questions and draft answers that can be reviewed and used to test Study Coach AI's intended functions. These include answering course-related questions, explaining concepts, adapting explanations to different levels of understanding, and indicating when an answer requires further verification.

It can also support preliminary evaluation of answer accuracy, grounding in approved course materials, source visibility, explanation quality, and error handling. However, because the dataset is small and synthetic, it cannot by itself demonstrate reliable performance across different subjects, student populations, or real-world learning situations.

The dataset will therefore serve as an initial prototype resource rather than a comprehensive or fully verified educational knowledge base.

4. Appendix A — Prompt Log: Revised Format

Purpose

This appendix documents the use of ChatGPT during the preparation of the Study Coach AI project. It records the prompts supplied to the tool, the outputs described in the project notes, and the team's review responsibilities.

Only prompts that were actually submitted should be classified as completed entries. The output summaries and revision notes must reflect what occurred during the actual interaction. Planned prompts are listed separately and must not be presented as completed activities.

Entry 1: Developing the Week 3 Deliverables

Tool Used: ChatGPT

Prompt Used:

"Act as a Human-AI Interaction (HAI) project assistant. Help me prepare the Week 3 deliverables for a project called Study Coach AI, an educational assistant designed to help students understand course materials, practice questions, and prepare for exams. Create a structured draft that includes three fictional user personas with different technical proficiency levels, bias-check notes, and a data-source description. Ensure that at least two personas are novice AI users, at least one persona has an accessibility need, and the personas represent diverse cultural backgrounds. Clearly distinguish proposed ideas from information that the project team still needs to verify."

Output Generated:

The documented output included a structured draft for three fictional personas, a bias-review framework, and a proposed description of a synthetic educational question-and-answer dataset.

Team Review and Changes:

The team compared the proposed personas with the assignment requirements and documented the intended distinctions between technical proficiency, AI experience, and accessibility needs. The dataset description was revised to distinguish the planned target of 30 records from the three sample records currently shown.

Any additional revisions should be recorded only after the team has completed them.

Reason for Using the Prompt:

To organize the Week 3 deliverables and establish an initial draft that could be reviewed against the instructor's requirements.

Entry 2: Reviewing and Organizing the Project Requirements

Tool Used: ChatGPT

Prompt Used:

"Act as a Human-AI Interaction (HAI) project reviewer. Review the Week 3 requirements for the Study Coach AI project and help organize the deliverables into a clear, structured document. Ensure that the document covers three fictional user personas with different technical proficiency levels, at least two novice AI users, at least one accessibility need, diverse cultural backgrounds, bias-check notes, and a data-source description. Also, identify the information that needs to be verified by the team before submission."

Output Generated:

The documented output provided an outline of the required Week 3 deliverables and identified details requiring team verification, including persona coverage, dataset status, and source permissions.

Team Review and Changes:

The outline was compared with the assignment instructions. The documentation was refined to separate design proposals from implemented features and to clarify that the dataset target does not establish the actual number of completed records.

The team must confirm any additional revisions before reporting them as completed.

Reason for Using the Prompt:

To check the completeness of the documentation and identify areas requiring clarification before submission.

Planned Entry 3: Reviewing Potential Biases

Status: Planned — not a completed tool interaction unless the prompt has actually been submitted.

Tool: ChatGPT

Planned Prompt:

"Evaluate the proposed Study Coach AI personas for potential cultural, linguistic, educational, technical-proficiency, and accessibility biases. Identify assumptions that could exclude or disadvantage certain students, explain why they may be problematic, and recommend specific improvements to make the personas more inclusive. Clearly distinguish potential risks from biases confirmed through testing."

Purpose:

To obtain an additional review of the persona assumptions and identify improvements for inclusive interaction design.

Documentation requirement: If used, record the actual output, the team's review, and the changes accepted or rejected.

Planned Entry 4: Reviewing the Dataset Design

Status: Planned — not a completed tool interaction unless the prompt has actually been submitted.

Tool: ChatGPT

Planned Prompt:

"Review the proposed synthetic educational question-and-answer dataset for Study Coach AI. Assess whether its planned fields support answer grounding, source traceability, and preliminary evaluation. Identify limitations of a small synthetic dataset and explain what licensing, copyright, privacy, and permission checks the team should complete before using external materials. Do not assume that the proposed dataset has already been created."

Purpose:

To improve the dataset specification and identify the checks required before the data is used in the prototype.

Documentation requirement: Record the actual output and any changes the team makes after reviewing it.

Planned Entry 5: Checking Assignment Completeness

Status: Planned — not a completed tool interaction unless the prompt has actually been submitted.

Tool: ChatGPT

Planned Prompt:

"Act as an HAI assignment reviewer. Evaluate my Week 3 documentation against these requirements: three fictional personas with different technical proficiency levels, at least two novice AI users, at least one accessibility need, diverse cultural backgrounds, bias-check notes, a data-source description, and an AI prompt log. Identify missing or unclear elements and provide a checklist of improvements. Do not assume that any proposed change has already been implemented."

Purpose:

To identify remaining gaps before submitting the Week 3 documentation.

Documentation requirement: Add the actual output and the team's final review only after the interaction has taken place.

Prompt Log Documentation Note

The prompt log distinguishes completed interactions from planned activities. The exact prompt text should be preserved, and output summaries should accurately represent the results generated by the tool.

The team remains responsible for checking the accuracy of the generated material, verifying the dataset's actual status and source permissions, and confirming whether proposed revisions were implemented. AI-generated drafts are treated as supporting material rather than as evidence that implementation or validation has occurred.

All examples and persona details used in AI-assisted development must remain fictional and must not contain identifiable personal information.